

GEOG 281A

Spatial Relationships and Topology

Figuring out how things relate to one another, formally

Dr. Peter Kedron



SPATIAL RELATIONSHIPS

NAÏVE GEOGRAPHY

Formal models of the common-sense geographic world as the basis for designing Geographic Information Systems

An approach based in the feedback loop between (1) the development of formal geographic models that can be implemented in software, and (2) the testing of those models and their match with human perspectives

SPATIAL RELATIONSHIPS

How do people reason about spatial relationships?

Quantitative or geometric (precise distance) v. Qualitative or topological (near, in)

Spatial relationships are of course presented using language, but in inconsistent ways that depend on scale and complicate the application of typical logical properties (e.g., transitivity).

When dealing with disjoint objects, people tend to use qualitative or topological descriptions.

CONSTRAINED RELATIONS

Different set of disjoint objects have different possible spatial relationships

What are the point-to-point relations?

What are point-to-line relations?

...

How can spatial relationships be represented in a formal geographic model that works within the point set topology of GIS? Does such a model match common human understand of these relationships?

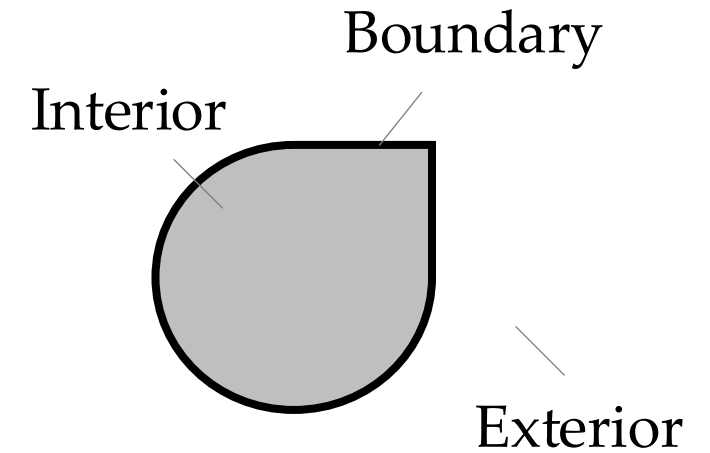
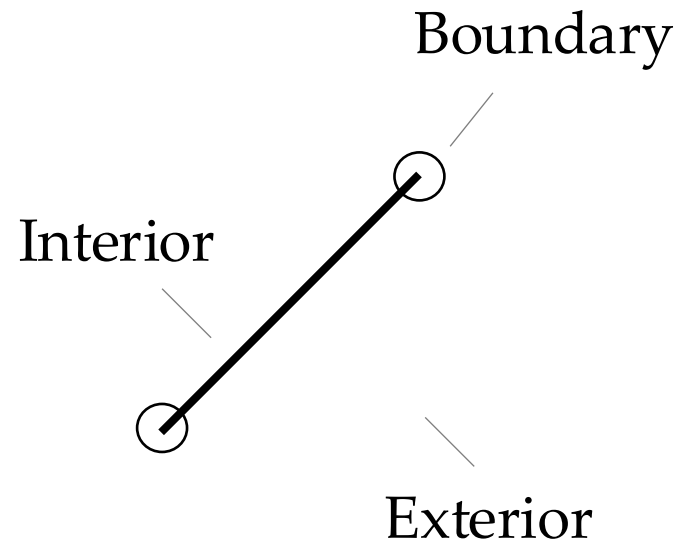
STUDENTS PERSPECTIVE

If we want to design GIS with Naive geography in mind, will GISystems require constant adaptations as social constructs evolve over time (and across cultures)?
Is this feasible?

Why assume there is one shared naive geography when people may conceptualize space differently?

THE NINE INTERSECTION MODEL

Interior, Boundary, and Exterior



INTERIOR

Space within the object

BOUNDARY

The edge of the object that separates the interior an exterior

EXTERIOR

Space outside the boundary of the object

THE DE-9IM

A 3x3 Matrix that outlines the intersection of these three elements

$$\text{DE9IM}(a, b) = \begin{bmatrix} \dim(I(a) \cap I(b)) & \dim(I(a) \cap B(b)) & \dim(I(a) \cap E(b)) \\ \dim(B(a) \cap I(b)) & \dim(B(a) \cap B(b)) & \dim(B(a) \cap E(b)) \\ \dim(E(a) \cap I(b)) & \dim(E(a) \cap B(b)) & \dim(E(a) \cap E(b)) \end{bmatrix}$$

THE DE-9IM

A 3x3 Matrix that outlines the intersection of these three elements

		Object B			Intersection
		INTERIOR	BOUNDARY	EXTERIOR	
Object A	INTERIOR				
	BOUNDARY				
	EXTERIOR				
					- Empty
					0 Point
					1 Line
					2 Area

ACTIVITY

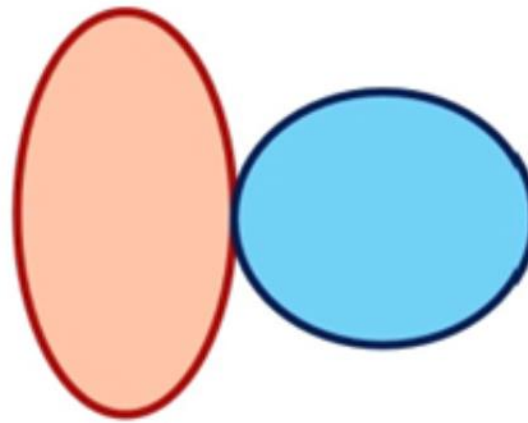
Purpose: To demonstrate that translating language into formal topological relations may be tricky

Task:

- Work in pairs, solve a topological riddle and represent it in a drawing
- All together, complete the DE-9IM bitmap for that riddle
- If the pair is wrong, receive topological ‘punishment’

TOPOLOGICAL RIDDLE I

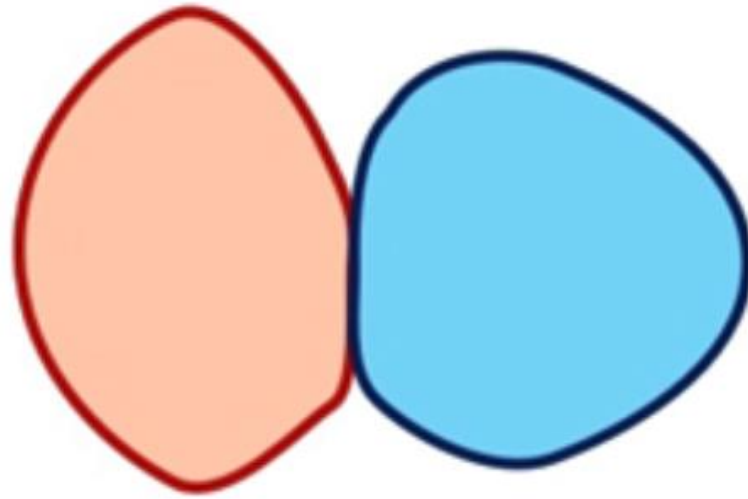
We are two very shy neighbors. We never enter each other's homes, and we like to keep our personal space. Most of the time, we stay completely separate. But every morning, when we both step outside to grab the newspaper, the corners of our fences awkwardly tap for just a split second.



$$\begin{pmatrix} - & - & 2 \\ - & 0 & 1 \\ 2 & 1 & 2 \end{pmatrix}$$

TOPOLOGICAL RIDDLE II

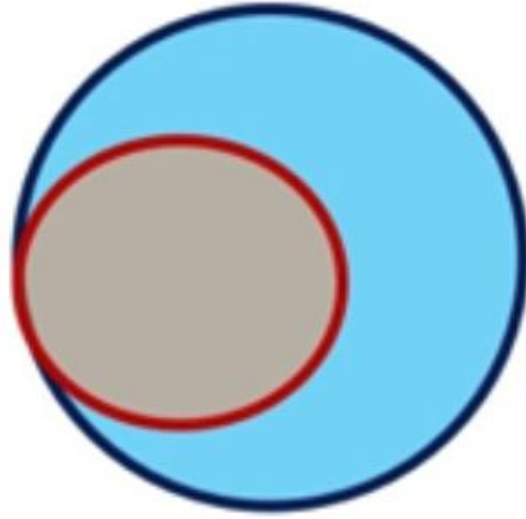
We are neighbors who get along surprisingly well. We respect each other's personal space and never enter each other's homes, but our fences run side-by-side for a long stretch. Every summer, we lean on the same fence and gossip for hours.



$$\begin{pmatrix} - & - & 2 \\ - & 1 & 1 \\ 2 & 1 & 2 \end{pmatrix}$$

TOPOLOGICAL RIDDLE III

One of us is a goldfish, and the other is a fishbowl. The goldfish spends its entire life inside the bowl and never escapes into the outside world. But the goldfish dreams of freedom. Every day, it swims to the glass, presses against it, and stares out into the world beyond the bowl.

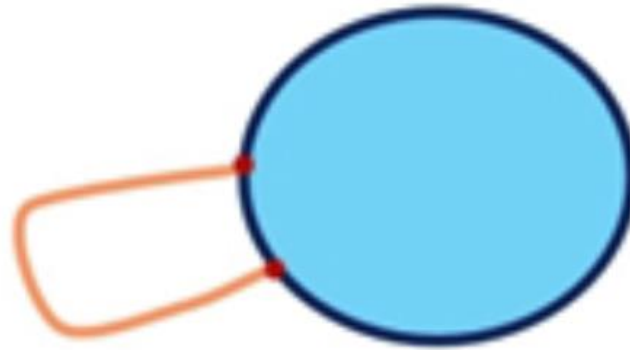


$$\begin{pmatrix} 2 & - & - \\ 1 & 0 & - \\ 2 & 1 & 2 \end{pmatrix}$$

TOPOLOGICAL RIDDLE IV

It is a rainy day. An angry senior professor is sitting inside their office with two closed doors.

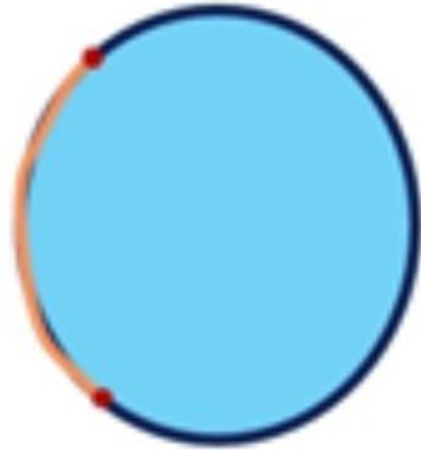
A stressed graduate student who desperately wants feedback on their research proposal has signed up for a meeting, but the professor appears to have forgotten. The student paces in the hallway leaving a muddy trail on the floor. They never enter the office, but leave mud marks on the doorstep as they place both hands on the doorframe, and try to listen for an improvement in the professor's mood.



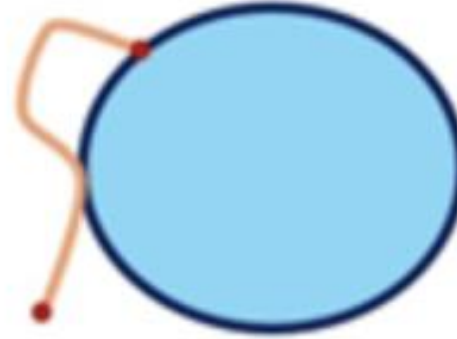
$$\begin{pmatrix} - & - & 1 \\ - & 0 & - \\ 2 & 1 & 2 \end{pmatrix}$$

TOPOLOGICAL RIDDLE V

The muddy shoed student waiting for the professor has now arrived five minutes late for class. They are trying very hard to sneak into the classroom without interrupting lecture. Instead of entering the room properly, the student awkwardly slides along the doorframe, constantly leaning against the wall, they attempt to melt into the wall.



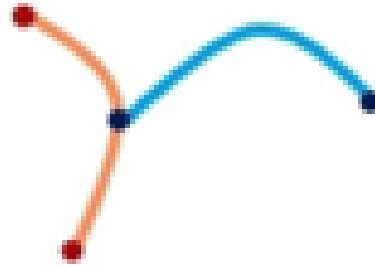
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$$\begin{pmatrix} - & 0 & 1 \\ - & 0 & 0 \\ 2 & 1 & 2 \end{pmatrix}$$

TOPOLOGICAL RIDDLE VI

Two old friends spot each other across campus and run toward one another for a dramatic high five. Unfortunately, both of them completely misjudge the timing. Instead of colliding body-to-body, only the very tips of their fingers make contact for one brief, awkward moment before they continue past each other in embarrassment. Their paths continue onward afterward.



$$\begin{pmatrix} - & 0 & 1 \\ - & - & 0 \\ 1 & 0 & 2 \end{pmatrix}$$

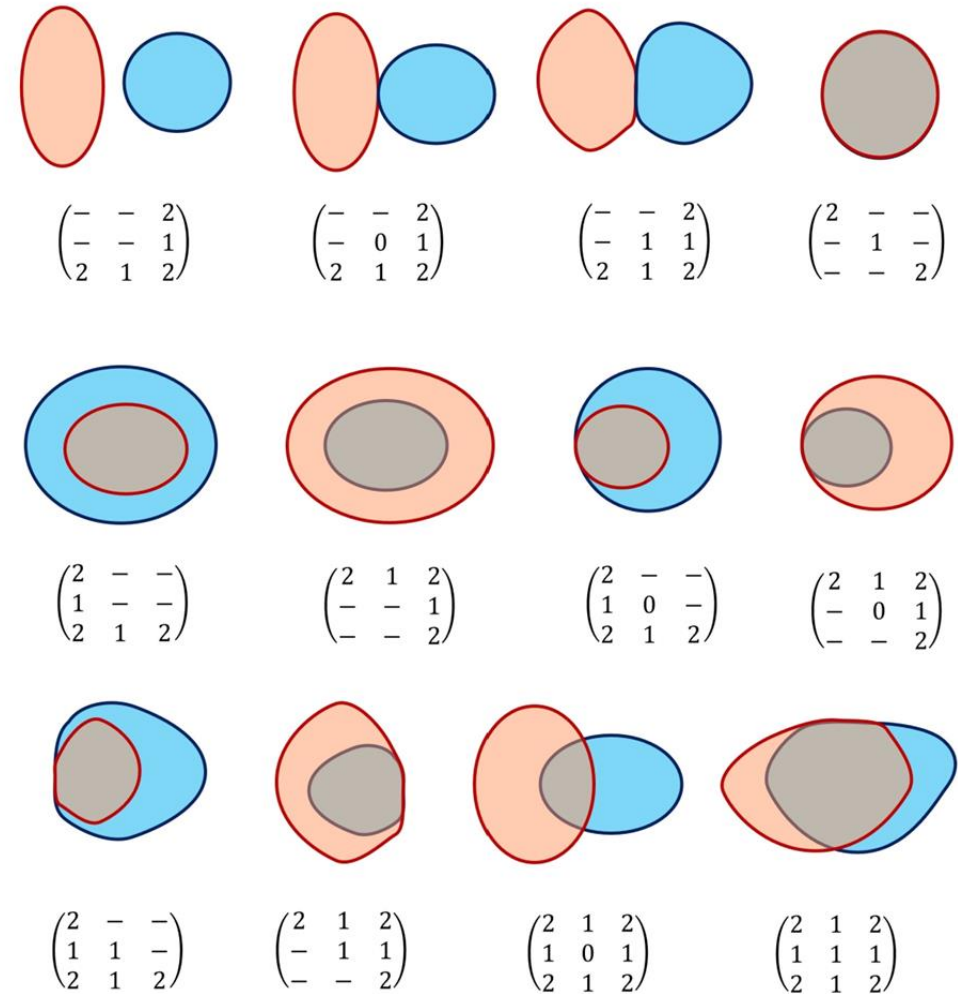
THE DE-9IM

There are

12 region-region intersections

31 line-region intersections

48 line-line intersections



BREAK

ACTIVITY

Purpose: Well, you'll see... Hopefully

Task:

“Here are sketches of a road and a state park. Please arrange the sketches into several groups, such that you would use the same verbal description for the spatial relationship between the road and the park for every sketch in each group.”

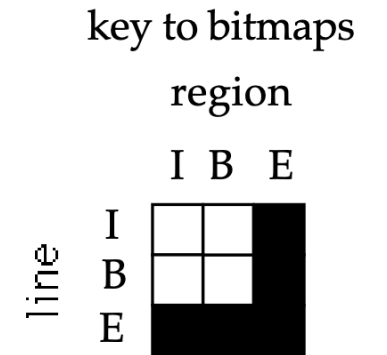
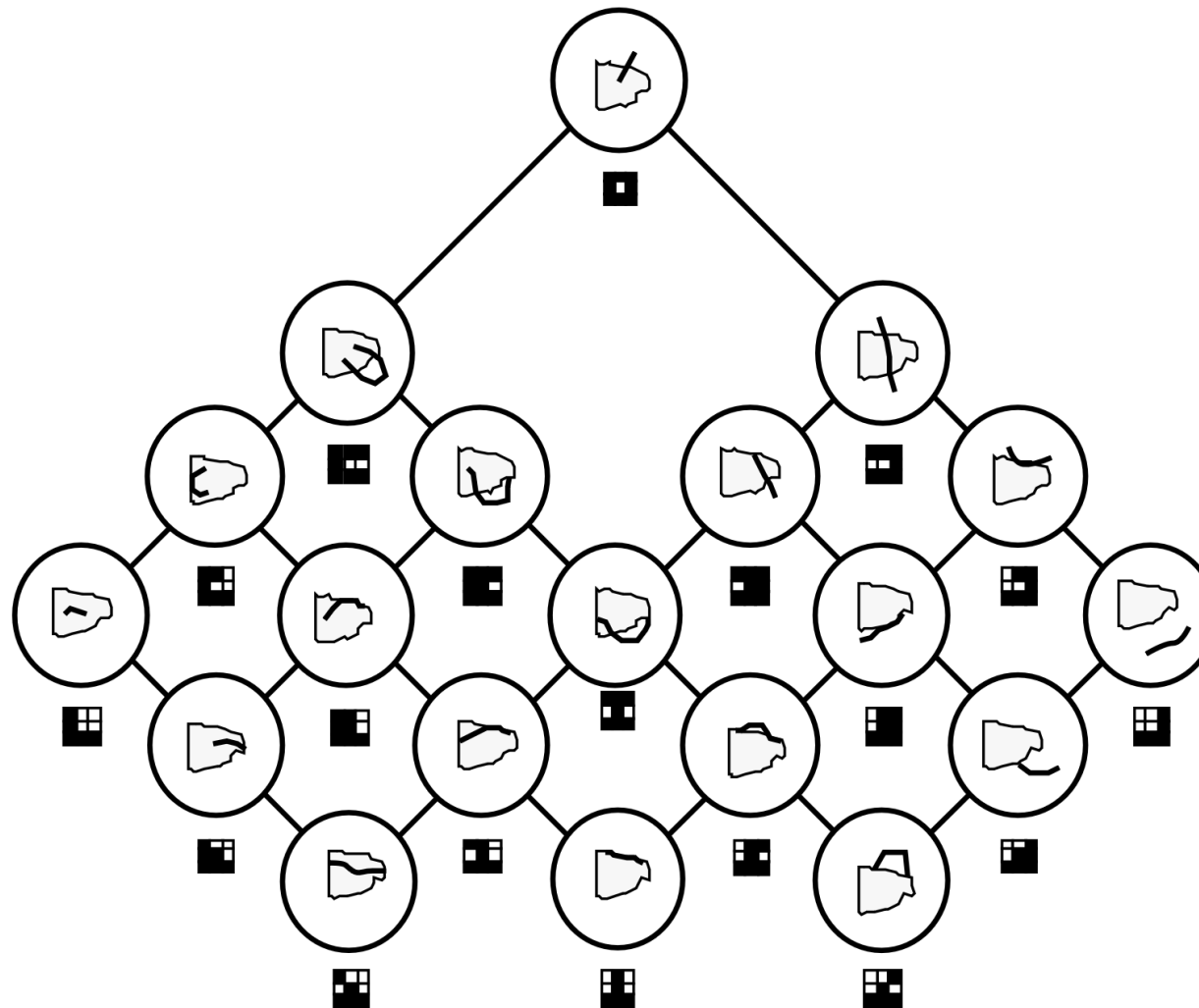
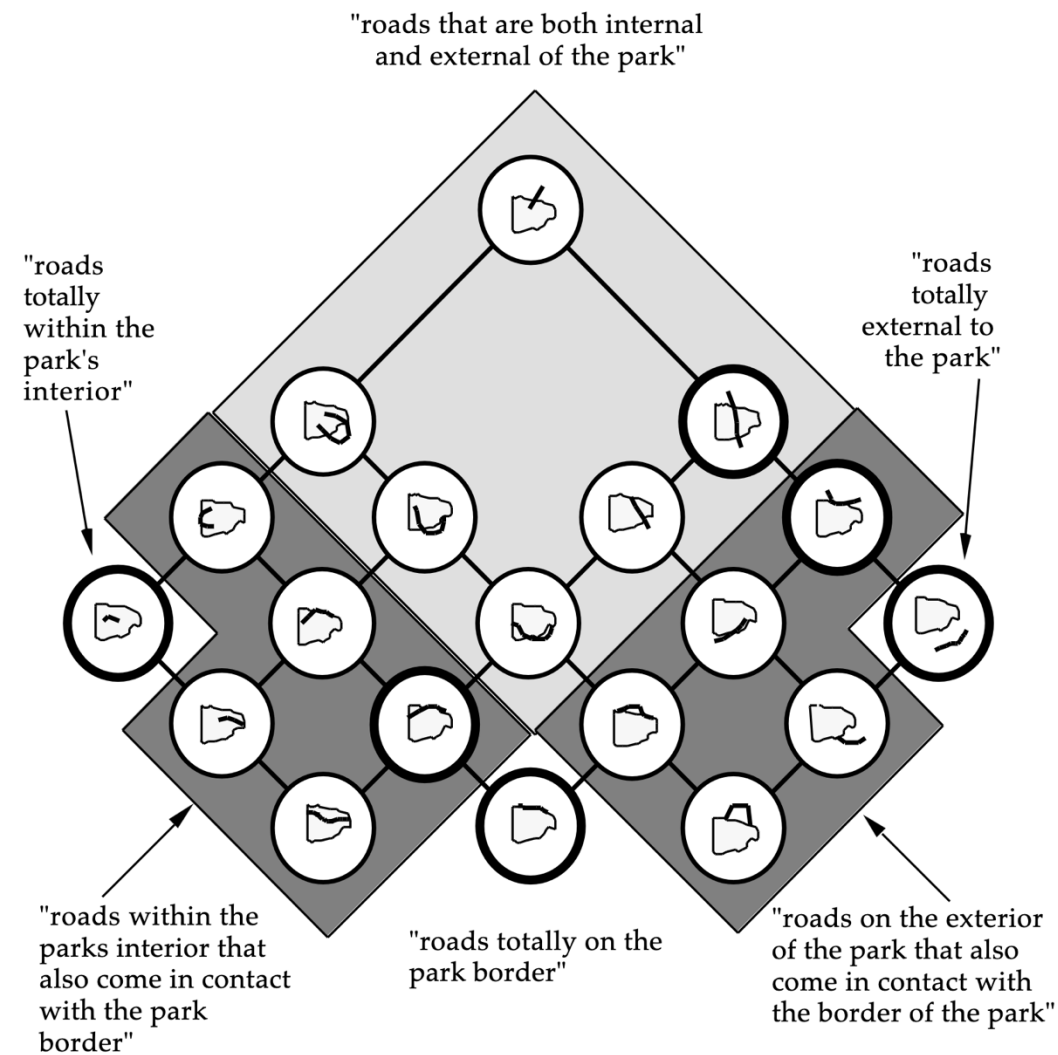
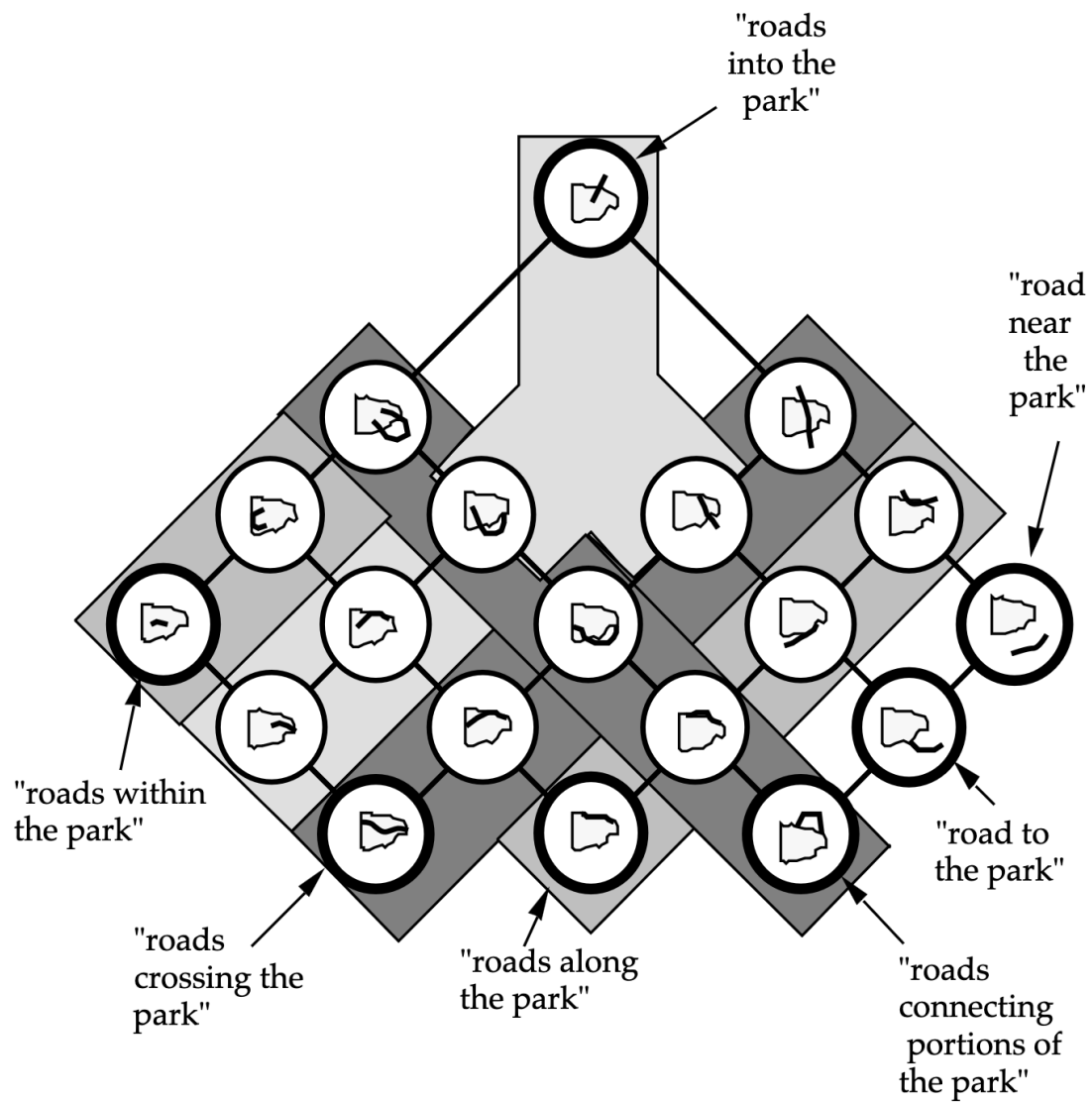


Figure 5. Diagram illustrating relationships among the 19 topologically-distinct spatial relations between a line feature and a region feature according to Egenhofer and Herring's "9-intersection" model. Situations are connected in the diagram if they differ for exactly one of the nine "intersections."



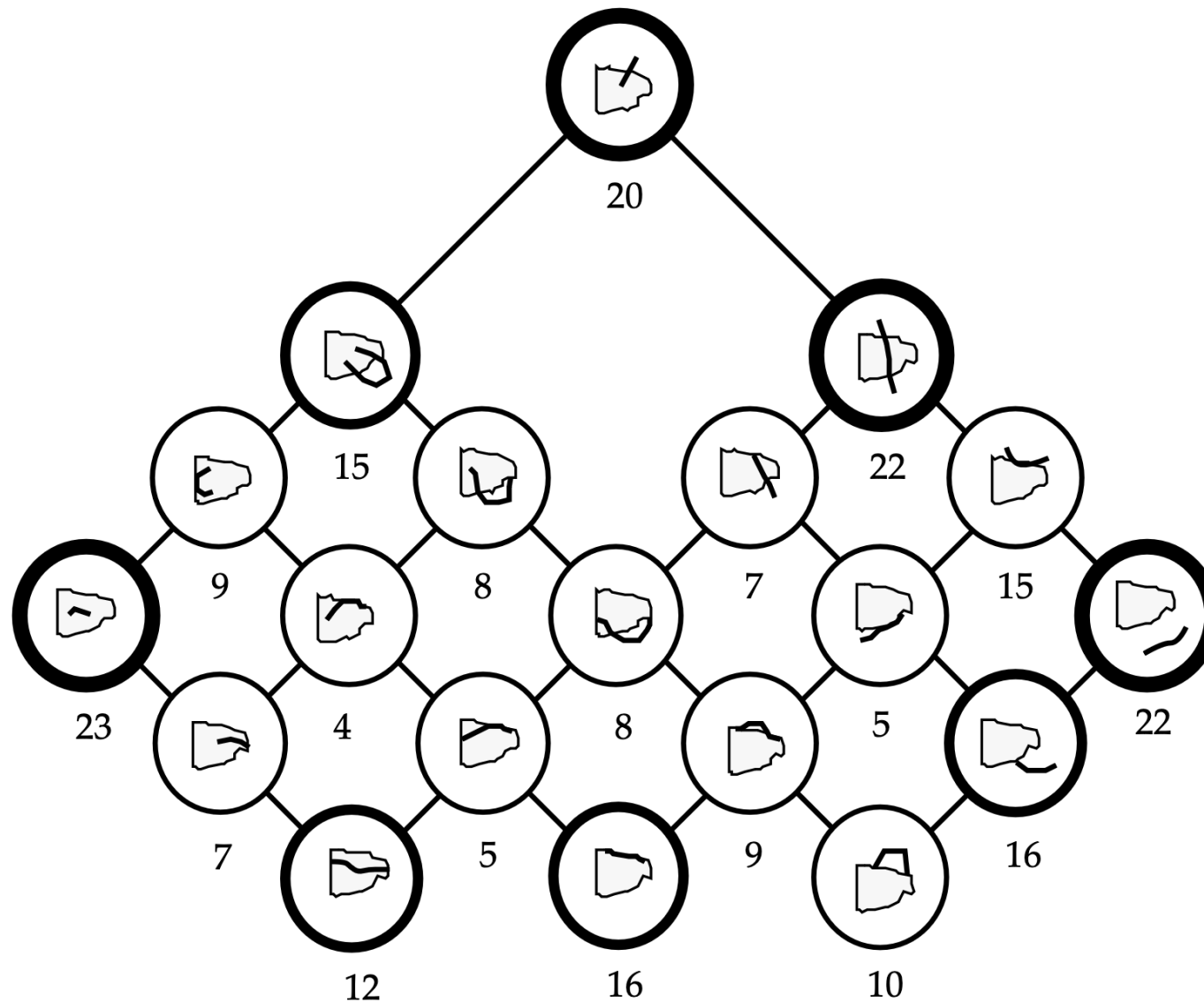
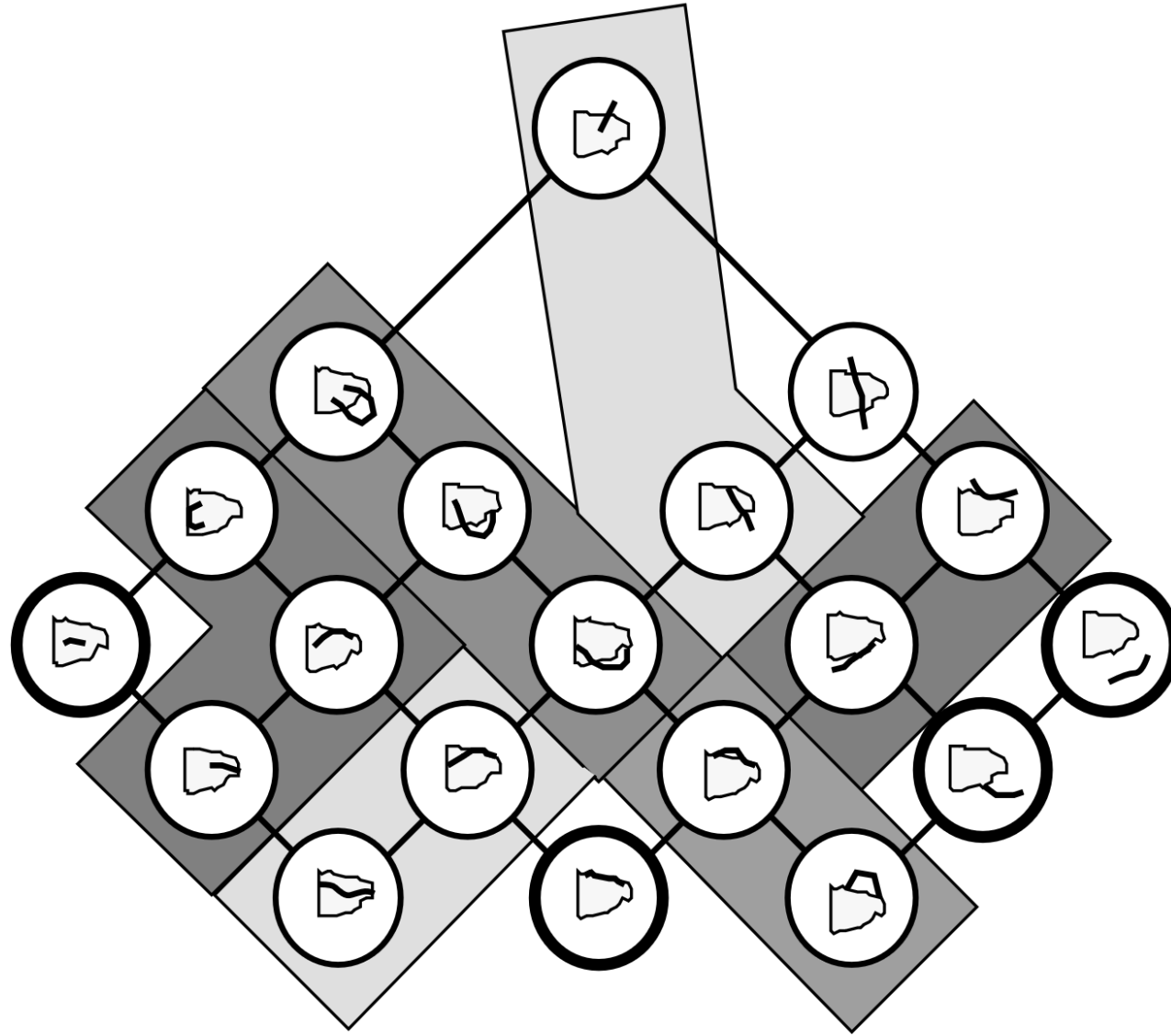


Figure 11. Frequency with which each situation was chosen as a group prototype by 23 subjects.



BREAK

SMITH (2001, p 131)

What sorts of entities are these, which can be brought into being simply by drawing lines on a map?

BOUNDARIES

- Bona Fide** Created by physical or material differences that exist independent of human perception.
- Fiat** Created by human decisions or decree and are dependent on human perception.

ACTIVITY

Purpose: To understand the difference between boundary types

Task:

- Use the 4 colors of stickers to identify each picture as a bona fide or fiat boundary.
- Make a note of the evidence or reasoning are you using to make you decision.

BOUNDARIES

Bona Fide and Fiat boundaries behave differently in the world, and interact differently with the point-set topology of the DE-9IM.

For Fiat objects, the boundary is part of the object it bounds and not a distinct point-set. This relation complicates the contact relationship between objects.

Smith argues that the logical need for fiat boundaries to share space when contacting limits topological relations to disjoint and touches

STUDENT PERSPECTIVE

Smith and Varzi argue that "fiat boundaries, and the analogue of contact which they involve, call for a different sort of topology which dispenses with the open/closed distinction" (p. 407), and further that fiat boundaries are oriented and lack the symmetry that bona fide boundaries enjoy (p. 415).

O'Sullivan distills the operational upshot: "possible relations among fiat objects are limited to being disjoint or touching" (Fig. 5.1, p. 123), adding that "it is unclear if any examples of genuine bona fide objects exist at geographical scales" (p. 122).

Yet OGC's DE-9IM ships all eight named predicates (Equals, Disjoint, Intersects, Touches, Crosses, Within, Contains, Overlaps) and computes them uniformly across every polygon in the database, whether it traces a coastline, a census tract, or a cadastral parcel that legally "actually are that thing" (O'Sullivan p. 143).

STUDENT PERSPECTIVE

If nearly every polygon we relate is fiat, then Within, Contains, Crosses, and Overlaps are being evaluated under a point-set regime Smith and Varzi say does not apply, and asymmetric fiat touches are being symmetrized by construction.

Is this a deep ontological mismatch GIS quietly papers over, with measurable downstream damage in adjudication, spatial joins, and overlay analytics, or is it a harmless idealization because the legal force of the map collapses the fiat/bona fide distinction the moment the polygon is drawn?

STUDENT PERSPECTIVE

If this mismatch has persisted despite decades of advances in GIScience, does it suggest a limitation of current computational frameworks themselves rather than simply a lack of better models or data?

How might GIScience change if ambiguity was treated as fundamental properties of geographic space rather than problems to be minimized?

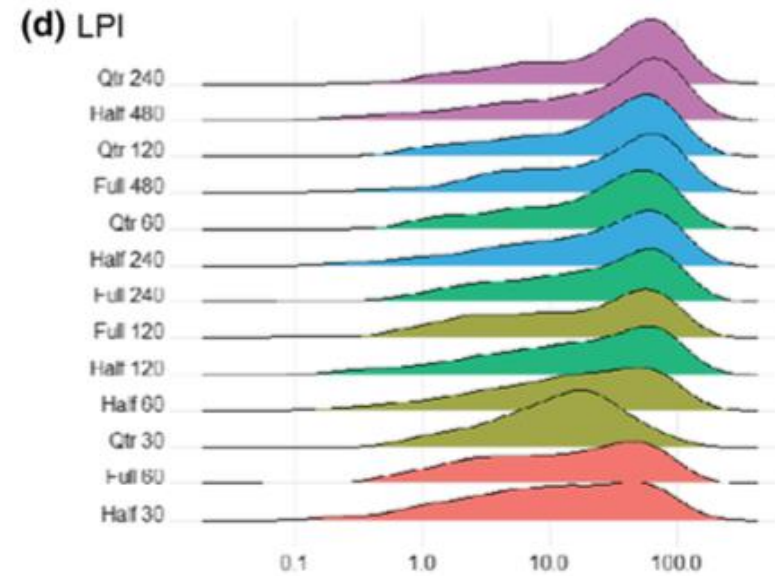
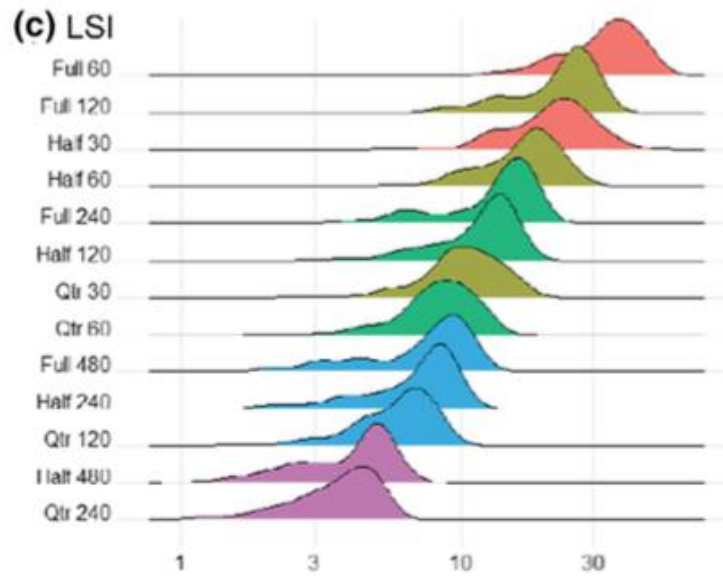
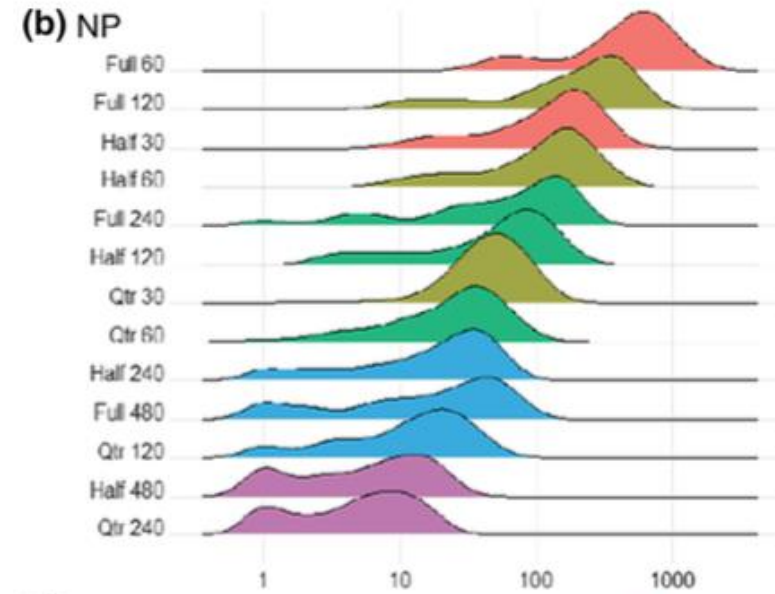
SCOPE

SCOPE

The dimensionless ratio between the range (extent) of a representation and the resolution (grain) of that representation

(a)

	Extent		
Grain km ² (res. m)	25 km ² (Qtr)	100 km ² (Half)	400 km ² (Full)
0.0009 (30 m)	B	A	
0.0036 (60 m)	C	B	A
0.0144 (120 m)	D	C	B
0.0576 (240 m)	E	D	C
0.2304 (480 m)		E	D



PRACTICAL ADVICE FROM DR FRAZIER

- FIRST** Researchers should set scope based on the phenomenon being studied, rather than relying solely on available data.
- SECOND** Studies should report both grain and extent quantitatively so that scope can be calculated and compared across studies.
- THIRD** Researchers should undertake comparisons and replications based on scope.

STUDENT PERSPECTIVE

Should spatial objects have the ability to preserve native resolutions after steps like aggregation? What implications would this have for replicability in research?